GlucoTwin

Personalized glucose forecasting between clinic visits

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Digital Twin Challenge 2026 · Happiest Health
Student research prototype — not a medical device

The problem: what happens between appointments

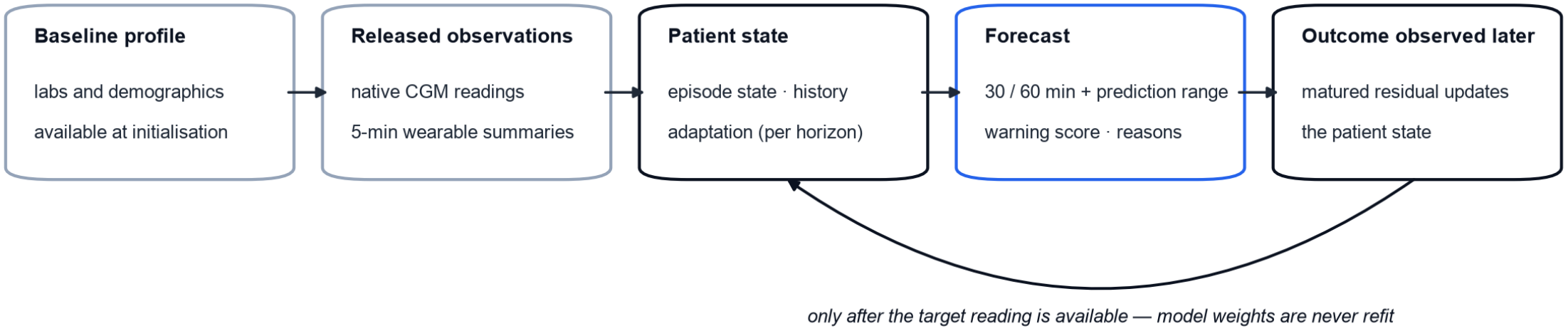
- HbA1c summarises ~3 months of glucose; it cannot show short high-glucose excursions (NIDDK).
- WHO identifies continuity of care and diabetes monitoring as ongoing challenges in India.
- Continuous glucose monitors produce ~288 readings a day — too many to review by hand.
- Question for the clinician: “Which monitored patient’s glucose pattern merits review, and what observations support that?”
- Our target: forecast glucose 30 and 60 minutes ahead and warn of a NEW sustained episode > 180 mg/dL (≥ 15 min), for Type 2 diabetes.

Use case and boundaries

- Hypothetical example (an illustration, not a deployment): a diabetes educator in Mumbai follows adults with Type 2 diabetes who wear CGMs between quarterly visits, and opens one person's replay to see forecasts, warnings and the data behind them.
- User: a clinician reviewing a small monitored cohort; secondary user: a reviewer inspecting evidence.
- Workflow (proposed): baseline labs at a visit → CGM + wearable measurements → twin forecasts → a review queue lists replays with unreviewed warnings first (an order of work, not triage) → each person reviewed in their own workspace.
- Shows: observed vs predicted glucose, uncertainty, data freshness, factual reasons, research warnings.
- Never: insulin or medication advice, triage, live device claims, LLM-generated predictions.
- Potential fit with diagnostics + outpatient services is a proposal — no partnership or pilot is claimed.

What makes it a twin (operational definition)

- Versioned patient state: baseline profile, episode state, adaptation state, issued forecasts.
- Incoming observations released by a replay clock (only what was available at that time).
- Population forecaster + patient-specific correction learned from that person’s matured forecast errors.
- Measured updates over time — a statistical research twin, not a physiological simulator.

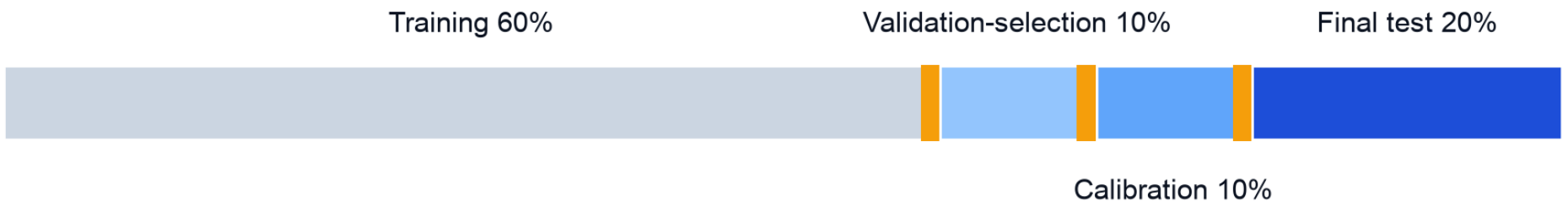


Data and provenance

- CGMacros 1.0.0 (PhysioNet, CC BY-NC-SA 4.0): 45 adults, ~10 days, Dexcom G6 Pro + Libre Pro (blinded), Fitbit, meals, screening labs.
- Primary cohort: 14 participants in the dataset's Type 2 group (HbA1c > 6.4 %); 31 others replayable but excluded.
- Finding: the published 1-minute CGM columns are linearly interpolated. We recover native 5-minute Dexcom readings (integer knots) and never count interpolated rows.
- Undocumented columns reported, not interpreted; documented error codes excluded; dates shifted; timezone unknown → synthetic clock, no time-of-day features.
- Synthetic fictional participants exercise edge cases (gaps, late meal logs, short history) with no download.

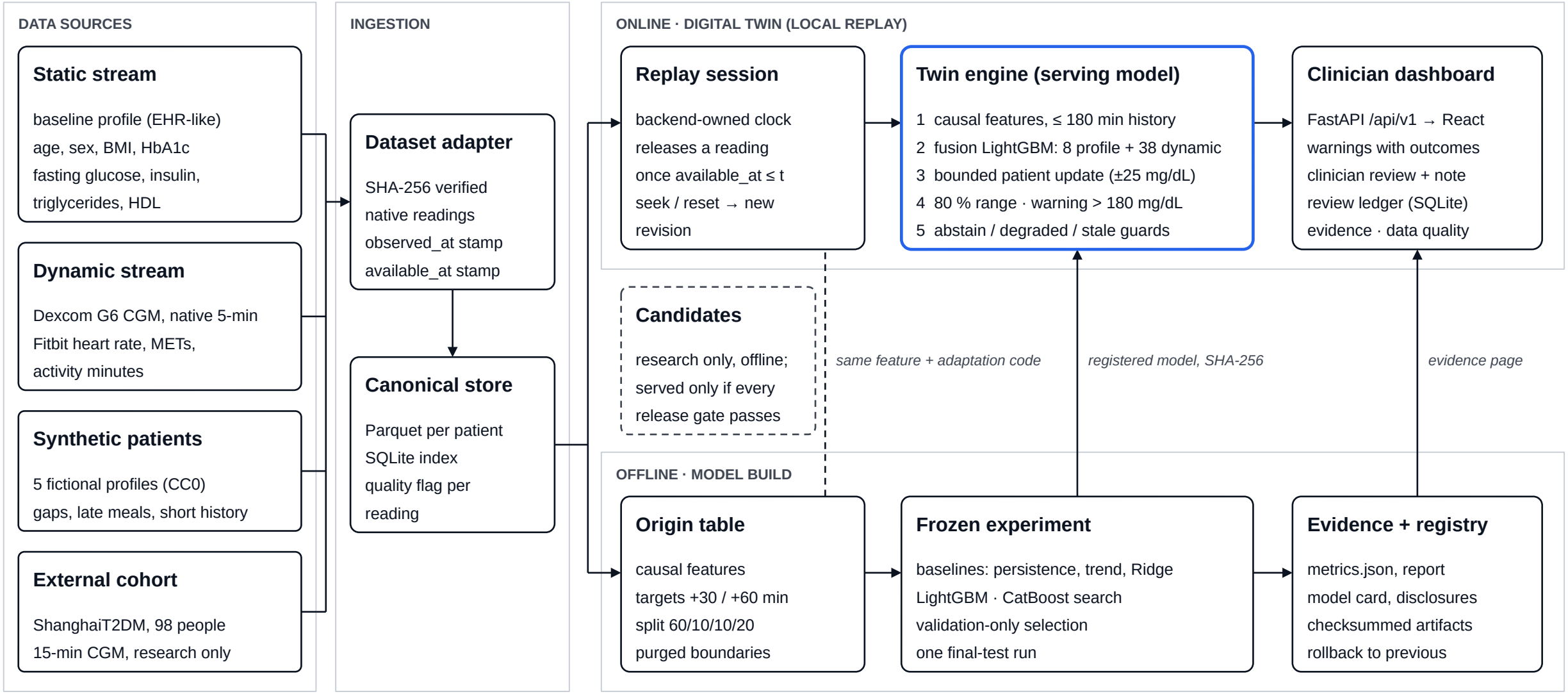
Information availability and leakage controls

- A forecast at time t uses only records observed and available by t.
- Meals left out of the main model (logging time unknown); tested separately.
- Protocol frozen before fitting; choices made on validation. The fusion re-run (gt-primary-2026-10-13-fusion) was declared after gt-primary-2026-10-11-v2's test had been inspected.
- The patient correction learns only from forecasts whose outcome was already observed.



One participant's timeline (~10 days); every participant is split the same way. Amber: purged origins whose 60-min target + 15-min confirmation would cross a boundary.

System architecture: two data streams, one twin



Solid arrow: data flow · Dashed line: identical code offline and online · Blue: serving model · Dashed box: research only, not served · CPU only, local, no cloud, no LLM · not a medical device

Models and personalization

- Fusion: 38 dynamic features (CGM, heart rate, activity) + 8 baseline profile values (age, sex, BMI, HbA1c, labs) in one model.
- Baselines: persistence, linear trend, ridge. Learners: LightGBM and CatBoost, same search budget.
- Patient correction: a bounded mean of this person's observed errors; model weights never refit.
- 80 % prediction ranges (split-conformal), calibrated for the exact serving pipeline.
- Warning score: highest forecast in the next 60 min minus 180 mg/dL — not a probability.

Why this learner: validation-only selection

Candidate	Val. MAE 30	Val. MAE 60	Score
lightgbm:dyn-prof-v2	**13.47**	**21.26**	**17.36**
catboost:dyn-prof-v2	13.56	21.31	17.44
lightgbm:dyn-v2	13.76	21.73	17.75
catboost:dyn-v2	13.88	21.66	17.77
ridge:dyn-v2	14.13	22.44	18.28
ridge:dyn-prof-v2	14.21	22.61	18.41
linear_trend:cgm-only-v1	15.66	26.33	20.99
persistence:cgm-only-v1	16.51	26.32	21.42

Rule: lowest mean per-participant validation MAE over both horizons; within 0.5 mg/dL → simpler model. Only dyn-prof-v2 (profile + wearable) was eligible, to meet the fusion requirement — a constraint added on 2026-10-13, after the test segment had been inspected. Selected: lightgbm:dyn-prof-v2 with patient correction. Without the constraint the rule picks lightgbm:dyn-v2 (gt-primary-2026-10-11-v2).

Safety guardrails built into the twin

- Abstains instead of guessing: no forecast with under 60 min of history or no new reading; marked Stale after 15 min; weak inputs flagged Degraded, with the reason.
- No input is used before it was available; the fusion model abstains when a health-record value is missing.
- Patient correction: only from observed outcomes, needs 6, capped at ±25 mg/dL, reset after a 3-hour gap.
- Warnings stay Outcome pending until they can be judged, then match an onset or count as a false alarm.
- 80 % prediction ranges shown with measured coverage; not a medical device, no treatment advice.
- Opened tests are locked; the app reproduced all 15,440 saved test forecasts within 0.05 mg/dL.

Results at a glance: three kinds of evidence, kept apart

Evaluated on	Model or procedure	MAE 30 / 60 min	Persistence 30 / 60
Serving model — later recordings of the same 14 CGMacros Type 2 people (a test over time, retrospective)			
Final 20 % of each recording	Fusion LightGBM + bounded patient correction (serving)	**12.96 / 20.26**	**14.84 / 23.23**
Participant-held-out research — 14 CGMacros people, each held out once (internal validation)			
Five participant folds (D-032)	CGM-only procedure (Ridge in 4 of 5 folds), not served	13.88 / 22.30	15.34 / 24.47
Same folds (D-053)	Peak-aware constrained fusion selection, not adopted	13.89 / 22.00	15.34 / 24.47
Same 14 people (D-054)	Zero-shot foundation model, CGM only, not served	14.55 / 22.92	15.34 / 24.47
External check — ShanghaiT2DM, 98 people, CGM only; D-045 was the cohort's second opening			
98 people, 15-min CGM (D-045)	Separate 15-min LightGBM, CGM only, not served	10.90 / 19.31	12.14 / 20.44

Mean absolute error in mg/dL (lower is better), averaged over participants. Each group is a different model and test — do not compare across groups. Serving ranges (target 80 %) covered 74.2 % / 74.0 %.

Research: a zero-shot foundation-model baseline (D-054)

- amazon/chronos-bolt-tiny (8.7 M parameters, apache-2.0), zero-shot on CGM only. Not fine-tuned, not a fusion model, not served.
- Vs carrying the last reading forward: -0.79 [-1.01, -0.59] mg/dL at 30 min, -1.54 [-1.81, -1.30] at 60 min — lower for 14 of 14 people.
- Vs our serving recipe refitted per held-out person: +0.42 [-0.70, +1.47] mg/dL at 60 min (22.92 vs 22.50) — not better; worse above 180 mg/dL.
- Same 14 previously inspected people: an internal comparison. Its pretraining data may include public CGM recordings.

Project outcomes

- Built: a working glucose twin — replay API, immutable forecasts, warnings with outcomes — and a clinician workspace.
- Found: the published 1-minute CGM values are interpolated; we use only native 5-minute readings.
- Later recordings of the same 14 people: MAE 12.96 / 20.26 mg/dL vs persistence 14.84 / 23.23; better for 14/14 people at 30 min and 13/14 at 60 min.
- Warnings: 61/64 onsets detected, 1.49 false alarms per observed patient-day; 48.4 % of all onsets warned ≥ 30 min ahead.
- Reproducible: synthetic mode needs no download; results regenerate from versioned configs.

Limitations

- 14 participants from one US site; no Indian participants.
- Blinded CGMs replayed with assumed zero latency; medication and meal timing unknown.
- Profile values are constant per person; no benefit shown for people held out entirely.
- Ranges under-cover (74.2 % / 74.0 % vs 80 % target); the warning threshold ignores review workload.
- Retrospective replay only — no clinical, workflow or safety study.
- Next: a prospective, consented Indian cohort and a full-fusion evaluation on new people.

Team, links, reproducibility and licenses

- Team Kryptonite — Avinash Gehi (team leader) · Hardik Hinduja. Vivekanand Education Society's Institute of Technology (VESIT), Mumbai.
- Code: public GitHub link — added after the team publishes the repository — MIT license (code only). Quick start: make setup · make synthetic · make dev.
- Demo video: unlisted YouTube link — added after recording and upload.
- Data: CGMacros 1.0.0 (PhysioNet, doi:10.13026/3z8q-x658, CC BY-NC-SA 4.0); ShanghaiT2DM (figshare, doi:10.6084/m9.figshare.21600933.v5, CC BY 4.0).
- Evidence: docs/judge_guide.md · reports/evidence_manifest.md · docs/submission/GlucoTwin_Research_Workbook.xlsx.
- Research workbook (every claim → its source): https://docs.google.com/spreadsheets/d/1u06FJSwKpWywDKrE9fGa2ZzivmY3eSLE/edit?usp=sharing